Exp 6: Monte Carlo Prediction and Control for Reinforcement Learning

Aim

To implement Monte Carlo Prediction and Control methods to estimate the value function and find the optimal policy using complete episodes of experience.

Algorithm

Monte Carlo Prediction:

1. Initialize the value function .
2. Generate a complete episode using a policy.
3. Calculate the return from each visited state.
4. Update the value function using the average return.
5. Repeat for multiple episodes.

Monte Carlo Control:

1. Initialize and a random policy.
2. Generate complete episodes using an -greedy policy.
3. Calculate returns for each state-action pair.
4. Update using the average return.
5. Improve the policy by choosing the action with maximum .
6. Repeat until the policy converges.

CODE

```
import numpy as np

import matplotlib.pyplot as plt

episodes = [

    [1, 2, 3],

    [2, 1, 4],

    [3, 2, 5],

    [4, 3, 6],

    [5, 4, 7]

]
```

```
gamma = 0.9
returns = []
for episode in episodes:
    G = 0
    for reward in reversed(episode):
        G = reward + gamma * G
    returns.append(G)
state_value = np.mean(returns)
print("Monte Carlo Returns")
for i, r in enumerate(returns):
    print(f"Episode {i+1}: {r:.2f}")
print("\nEstimated State Value:", round(state_value, 2))
plt.figure(figsize=(7,5))
plt.plot(range(1, len(returns)+1), returns, marker='o', label='Episode Return')
plt.axhline(state_value, linestyle='--', label='Average Return')
plt.title("Monte Carlo Prediction")
plt.xlabel("Episodes")
plt.ylabel("Return")
plt.legend()
plt.grid(True)
plt.show()
```

OUTPUT

Monte Carlo Returns

Episode 1: 5.23

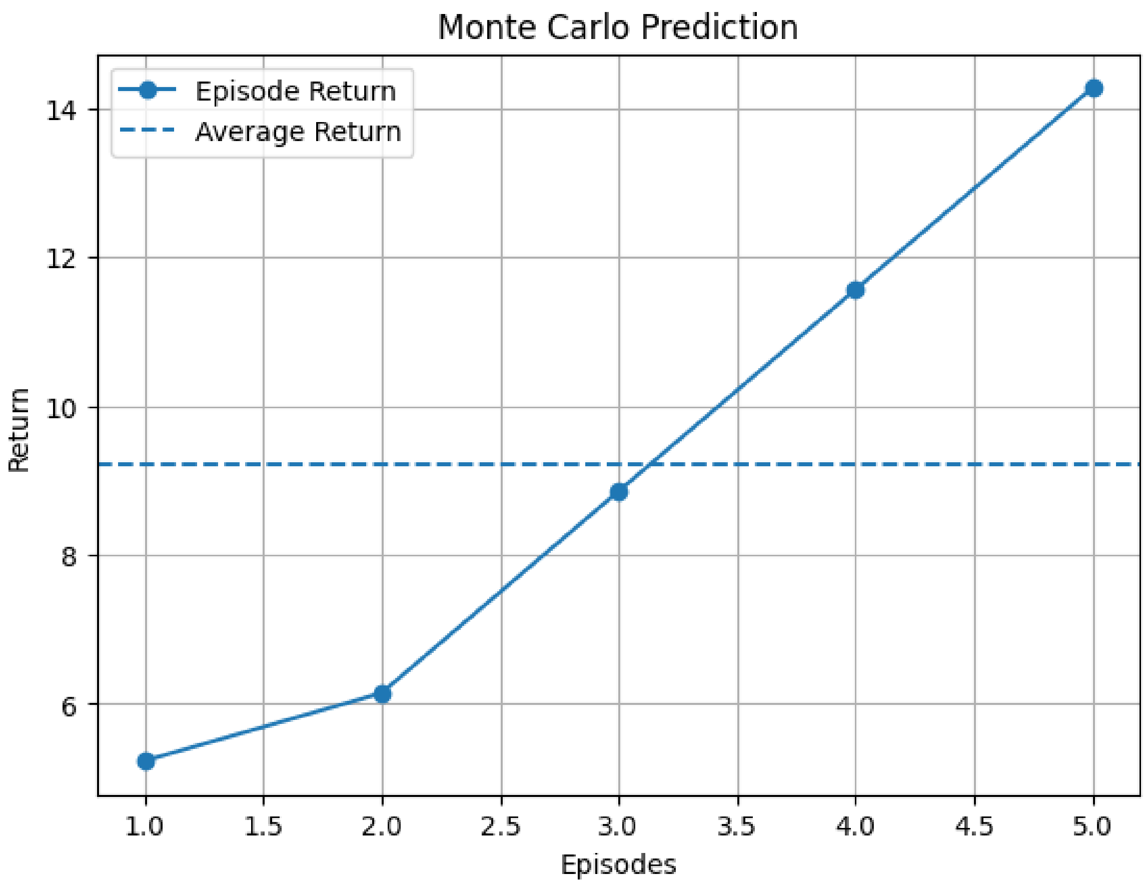
Episode 2: 6.14

Episode 3: 8.85

Episode 4: 11.56

Episode 5: 14.27

Estimated State Value: 9.21



Result

Thus, **Monte Carlo Prediction and Control** were successfully implemented. The prediction method estimated the value of states from complete episodes, while the control method improved the policy and identified the **best action for each state**.